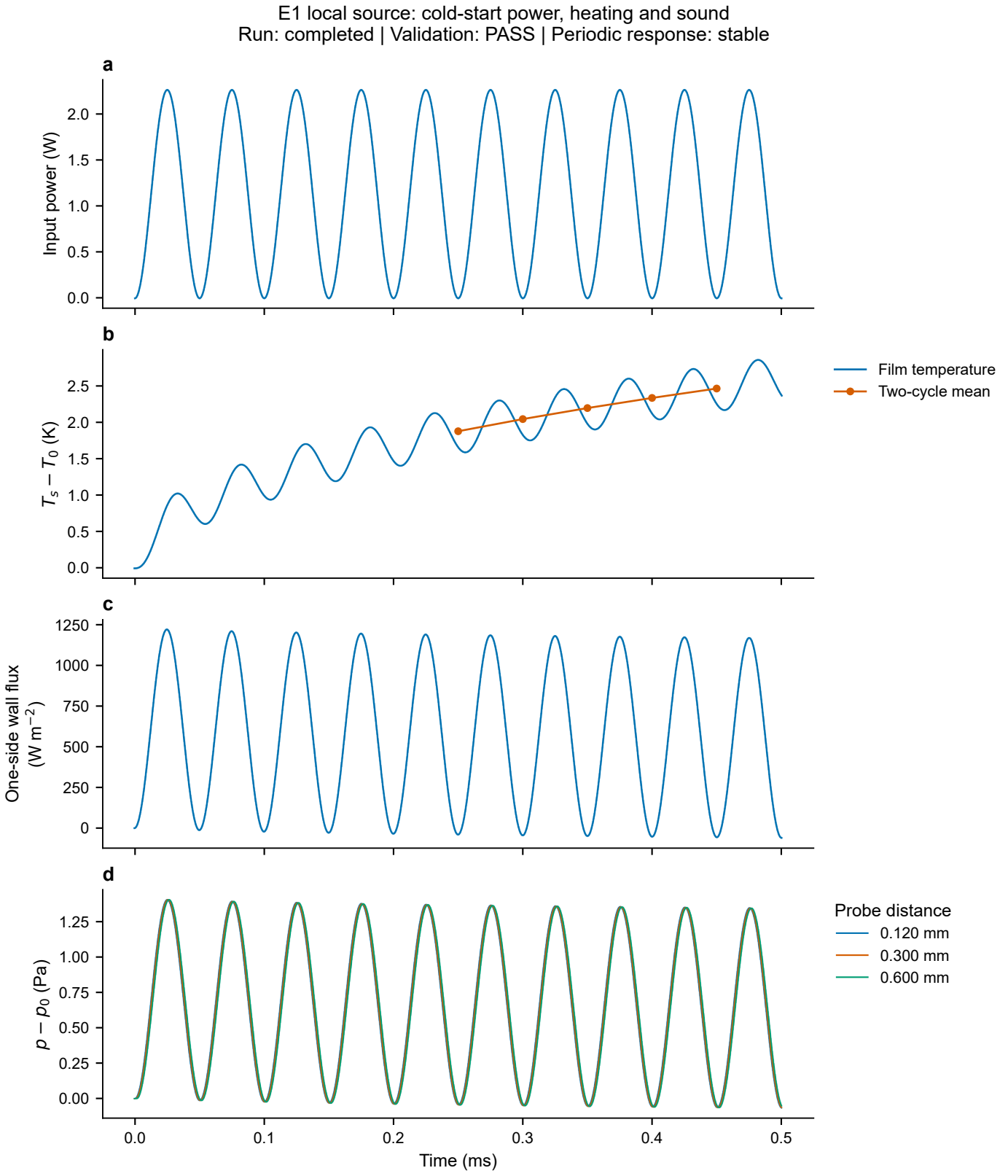
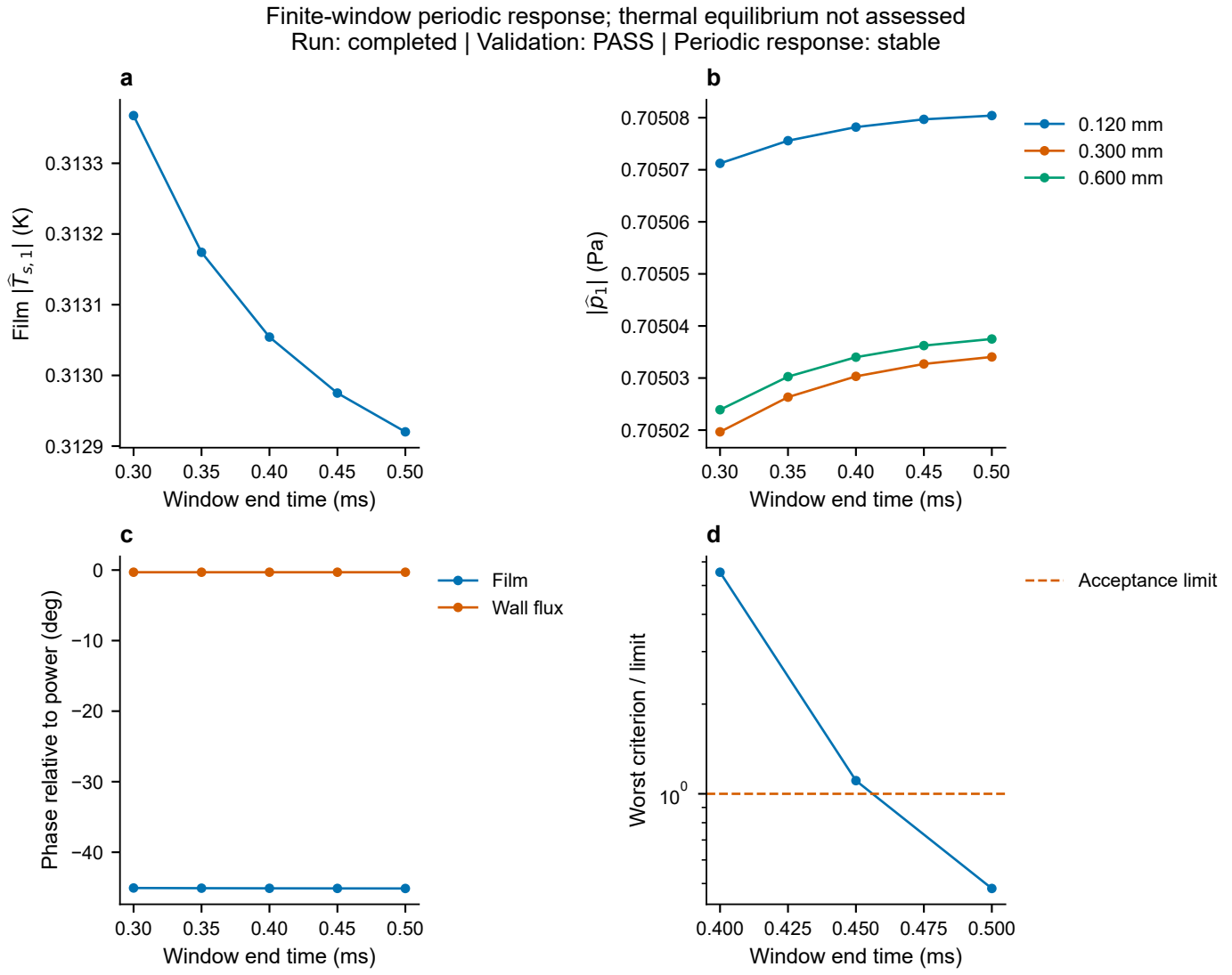
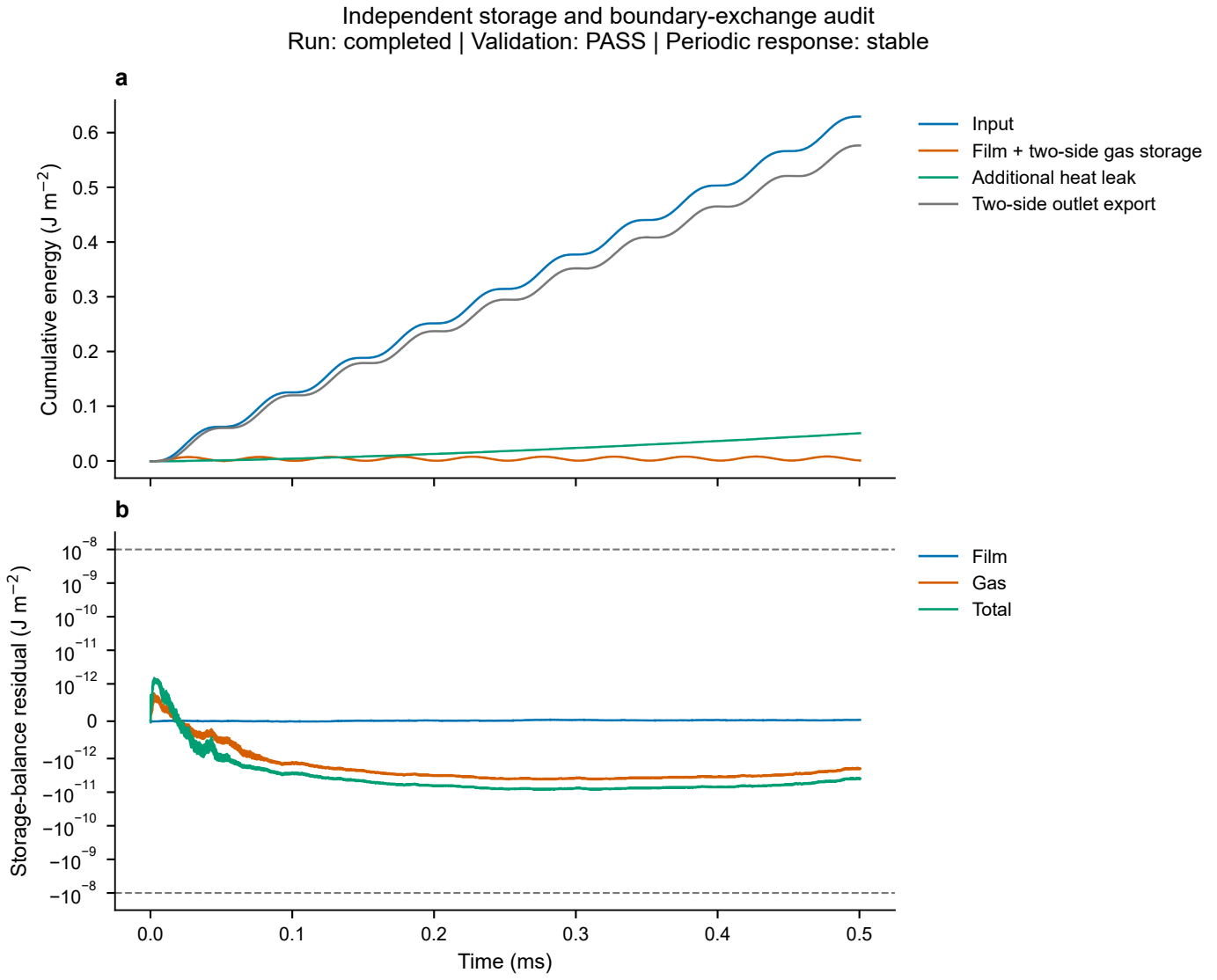
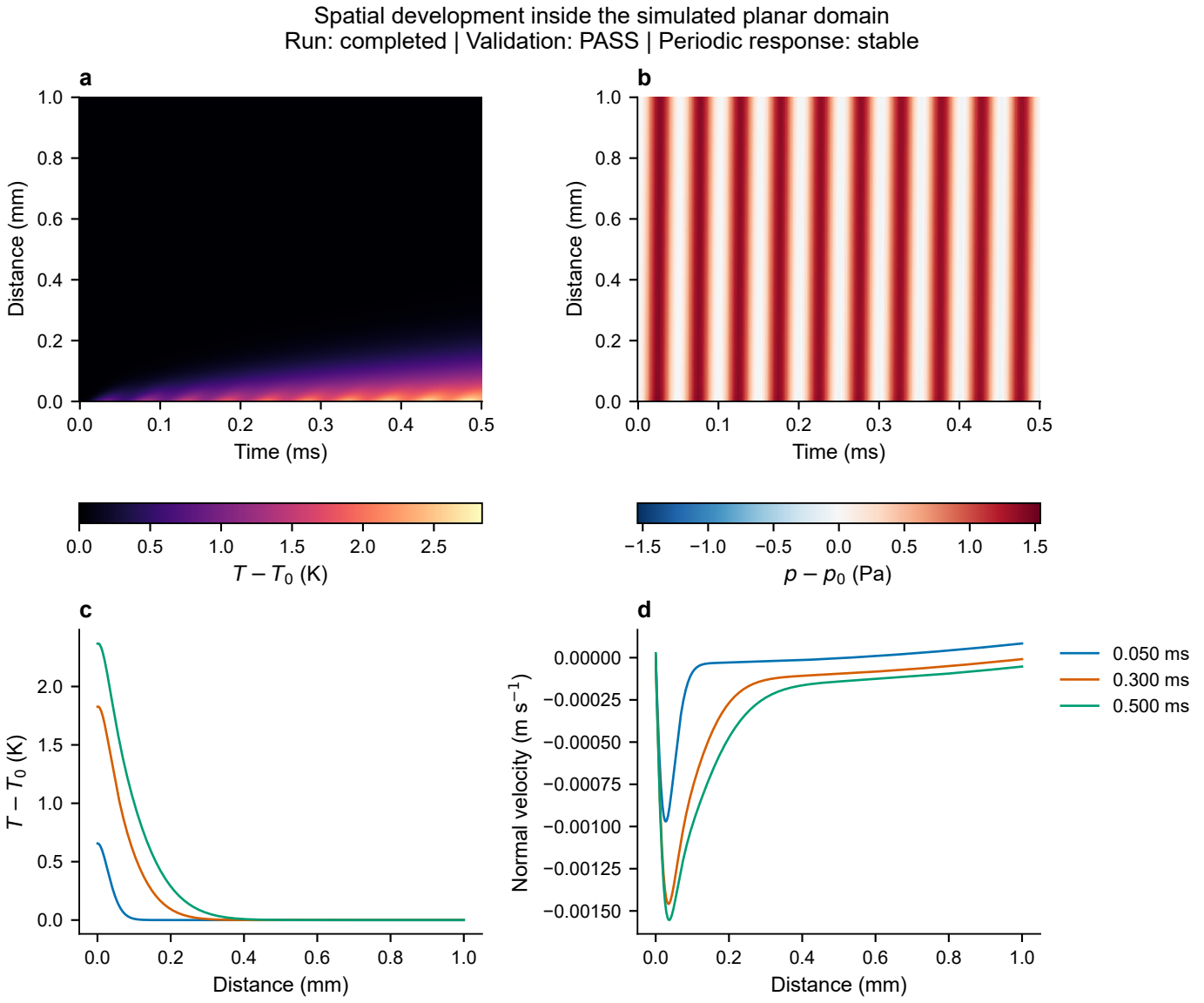

E1 local source: cold-start power, heating and sound
Run: completed | Validation: PASS | Periodic response: stable
a
Input power (W)
0.0
0.5
1.0
1.5
2.0
b
$T_s - T_0$ (K)
0.0
0.5
1.0
1.5
2.0
2.5
Film temperature
Two-cycle mean
c
One-side wall flux (W m^{-2})
0
250
500
750
1000
1250
d
$p - p_0$ (Pa)
0.00
0.25
0.50
0.75
1.00
1.25
Probe distance
0.120 mm
0.300 mm
0.600 mm
0.0
0.1
0.2
0.3
0.4
0.5
Time (ms)


Finite-window periodic response; thermal equilibrium not assessed
Run: completed | Validation: PASS | Periodic response: stable
a
Film $|\hat{T}_{s,1}|$ (K)
0.3133
0.3132
0.3131
0.3130
0.3129
0.30
0.35
0.40
0.45
0.50
Window end time (ms)
b
$|\hat{p}_1|$ (Pa)
0.70508
0.70507
0.70506
0.70505
0.70504
0.70503
0.70502
0.30
0.35
0.40
0.45
0.50
Window end time (ms)
0.120 mm
0.300 mm
0.600 mm
c
Phase relative to power (deg)
0
−10
−20
−30
−40
0.30
0.35
0.40
0.45
0.50
Window end time (ms)
Film
Wall flux
d
Worst criterion / limit
10^0
0.400
0.425
0.450
0.475
0.500
Window end time (ms)
Acceptance limit


Independent storage and boundary-exchange audit
Run: completed | Validation: PASS | Periodic response: stable
a
Cumulative energy (J m^{-2})
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Input
Film + two-side gas storage
Additional heat leak
Two-side outlet export
b
Storage-balance residual (J m^{-2})
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
0
-10^{-12}
-10^{-11}
-10^{-10}
-10^{-9}
-10^{-8}
Film
Gas
Total
0.0
0.1
0.2
0.3
0.4
0.5
Time (ms)


Spatial development inside the simulated planar domain
Run: completed | Validation: PASS | Periodic response: stable
a
b
Distance (mm)
Time (ms)
$T - T_0$ (K)
$p - p_0$ (Pa)
c
d
$T - T_0$ (K)
Normal velocity (m s^{-1})
Distance (mm)
0.050 ms
0.300 ms
0.500 ms